

SAINT LOUIS UNIVERSITY

International Student Compliance Analytics

End-to-End Analytical Journey | Week 1–4 Final Presentation

Team 5 | Muneeza Ali, Ambreen Zafar, Sanskriti Gupta
| April 2026

 n = 34,341 Students



PRESENTATION AGENDA

01

Dataset Overview & Compliance Context

Scope, sources, and compliance framework

02

Data Preparation & Validation

Cleaning, normalization, and quality checks (Week 1)

03

Exploratory Data Insights

Trends, distributions, and anomalies (Week 2)

04

Dashboard Walkthrough

Looker Studio build, KPIs, and validation logic (Week 3)

05

Recommendations

Eligibility, document compliance, and fee monitoring

06

Conclusion & Institutional Impact

Key takeaways and next steps

01 | DATASET OVERVIEW & COMPLIANCE CONTEXT

Data Sources

Connect Dataset

34,341 applicant records — enrollment, deposits, advisor assignments

SEVIS Dataset

I-20 issuance, SEVIS linkage status, government fee payment records

Data Source

Supabase PostgreSQL — refreshed 22 April 2026

Compliance Framework



I-20 Issuance



SEVIS Fee Payment



Enrollment Deposit



Advisor Assignment



SEVIS Linkage

Total Records Analyzed: 34,341 International Student Applicants

02 | DATA PREPARATION & VALIDATION — WEEK 1

1

Data Ingestion

Imported Connect and SEVIS CSV datasets into Supabase PostgreSQL. Verified row counts matched expected totals (34,341 records).

2

Column Normalization

Standardized date formats, trimmed whitespace, and enforced consistent naming across intake period fields (Fall/Spring/Summer).

3

Null & Duplicate Check

Identified null values in I-20 status and SEVIS linkage fields. Flagged duplicate applicant IDs for removal before analysis.

4

Compliance Field Validation

Validated binary flags for deposit status, fee payment, and advisor assignment. Confirmed referential integrity between Connect and SEVIS.

5

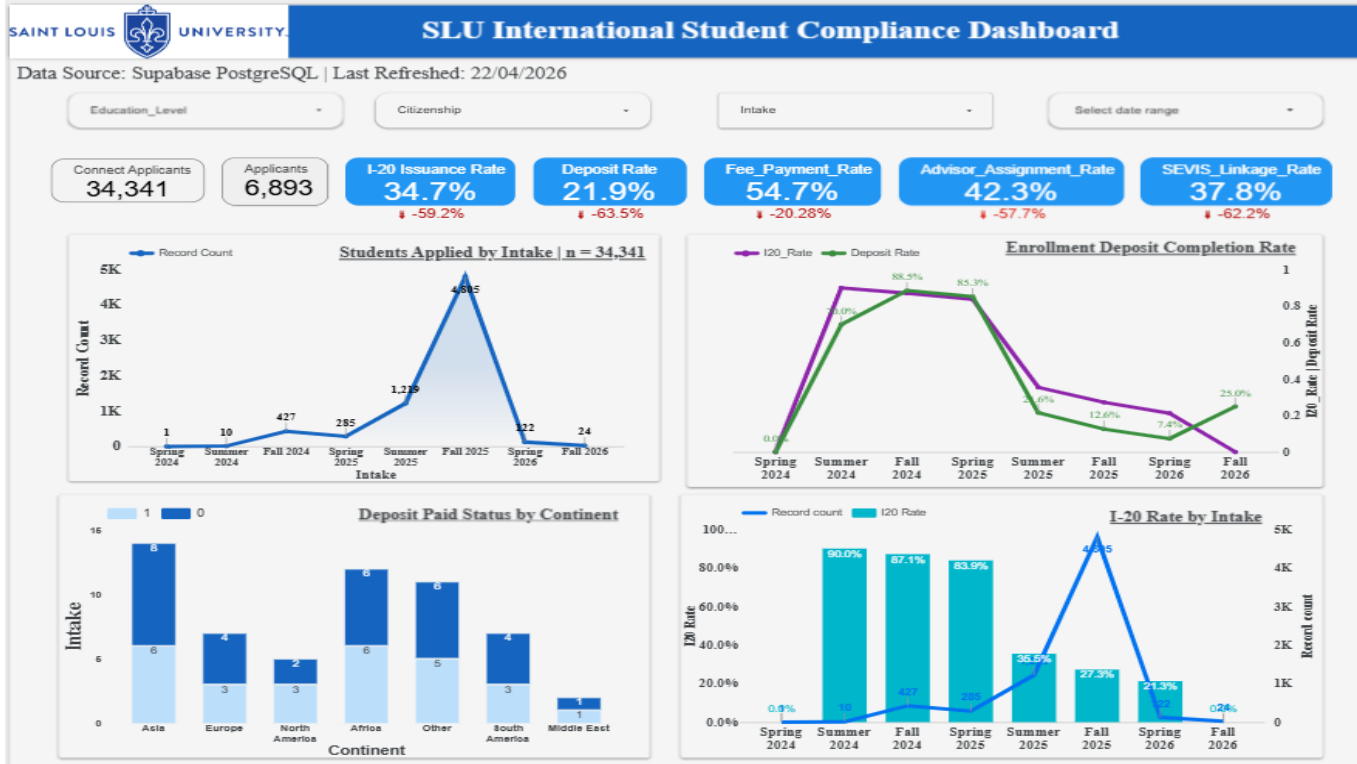
Data Dictionary

Documented each field's definition, data type, valid range, and compliance significance for team alignment and future audits.

6

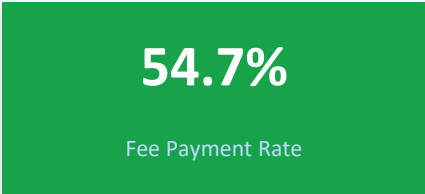
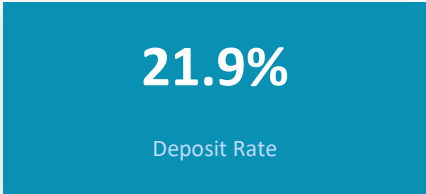
PostgreSQL Load

Final clean datasets uploaded to Supabase and verified. Tables linked via student ID for cross-dataset compliance queries.

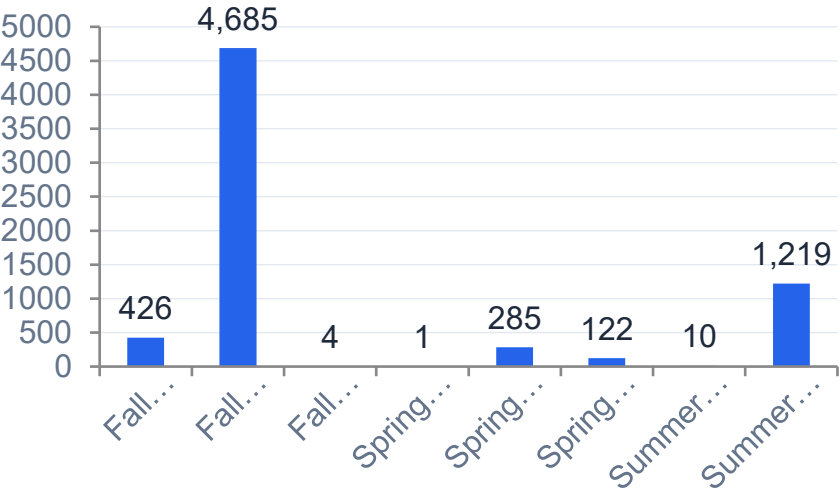


- Video Presentation Link: https://drive.google.com/file/d/1-kccmavZUVPzJ1cotaTfWYApV6nOdW-N/view?usp=drive_link
- Live Dashboard Link: <https://datastudio.google.com/s/l0fCNxgo0Ds>

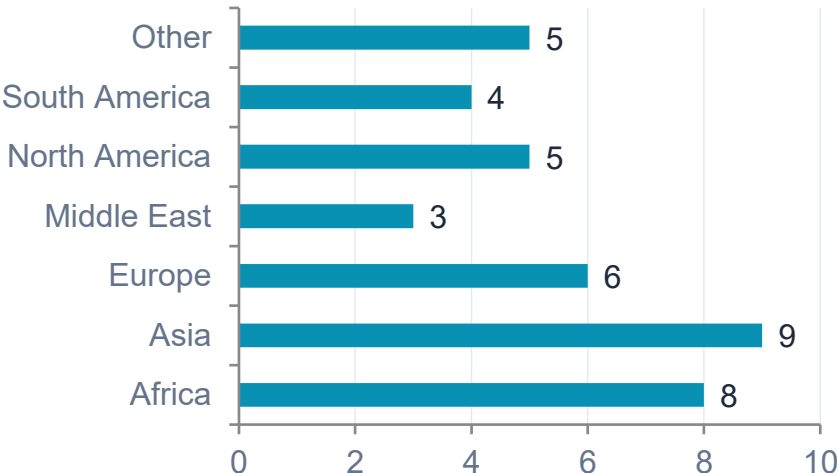
03 | EXPLORATORY DATA INSIGHTS — WEEK 2



Applications by Intake Period



Deposit Completion by Continent



03 | KEY FINDINGS FROM EXPLORATORY ANALYSIS

Declining I-20 Rates Across Intakes



I-20 issuance rate dropped from 88.1% (Fall 2024) to as low as 0.8% in Spring 2025, then partially recovered. Signals systemic processing delays or eligibility gaps.

Summer 2025 Shows Strong Recovery



Summer 2025 showed 35.5% I-20 rate with 1,219 applicants — the largest non-Fall intake. This cohort warrants focused monitoring for deposit and fee completion.

Deposit Rate Critically Low at 21.9%



Across all intakes, only 1 in 5 applicants completes enrollment deposit. Africa and Asia show highest volumes but still significant non-completion.

SEVIS Linkage Rate at 37.8% — High Risk



Over 60% of students lack confirmed SEVIS linkage. Without this, DHS compliance cannot be verified — posing institutional legal and audit risk.

Advisor Assignment Gap (42.3%)



Less than half of applicants have advisor assignments. Unassigned students are more likely to miss compliance milestones without proactive outreach.

All KPIs Show Negative YoY Trend



Dashboard shows: I-20 -59.2%, Deposit -63.5%, Fee -20.28%, Advisor -57.7%, SEVIS -62.2% — all declining. Structural process issues are likely, not isolated events.

Looker Studio Dashboard

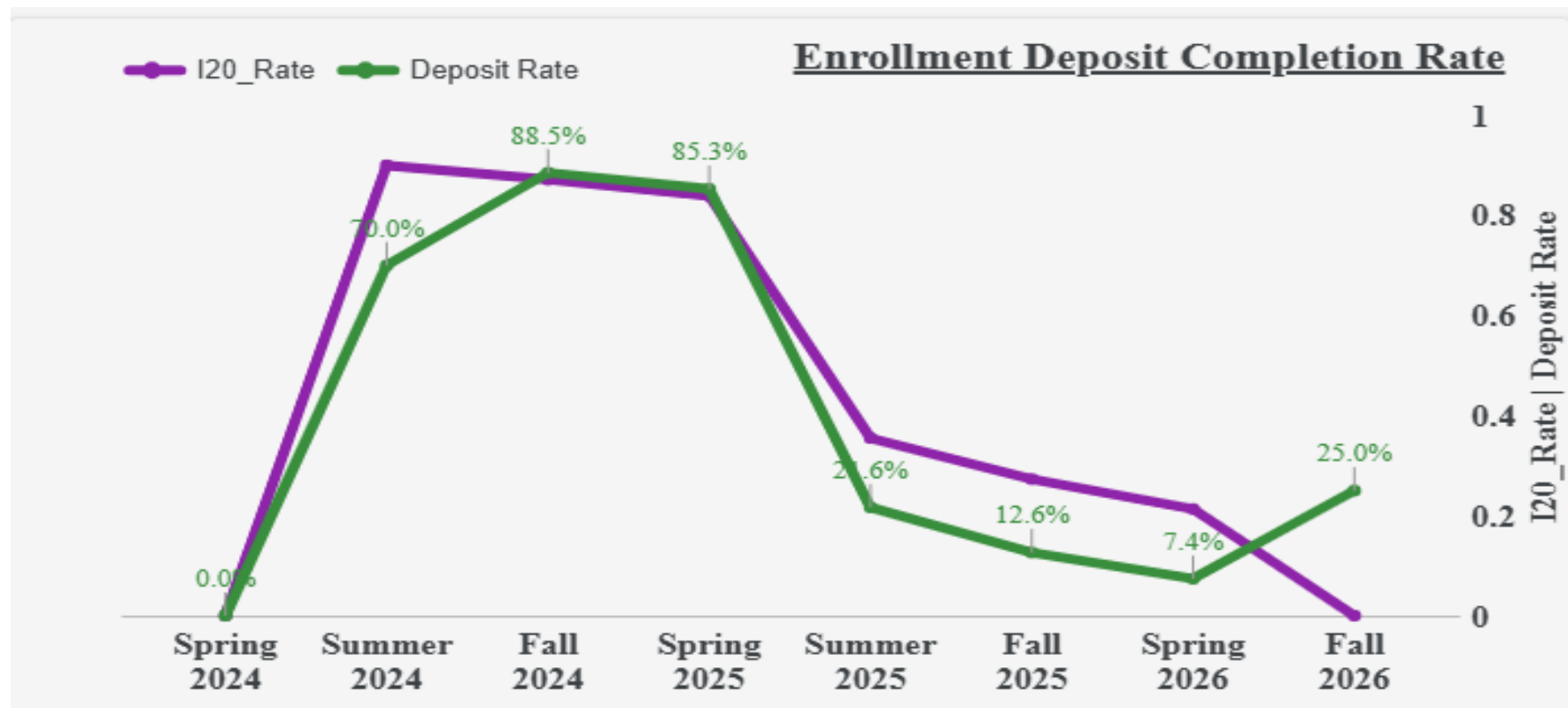
Platform:	Google Looker Studio (Data Studio)
Data Backend:	Supabase PostgreSQL
Last Refreshed:	22 April 2026
Total Records:	34,341 student records
Filters Available:	Education Level, Citizenship, Intake, Date Range

Key Performance Indicators

I-20 Issuance Rate	34.7%	-59.2%
Deposit Rate	21.9%	-63.5%
Fee Payment Rate	54.7%	-20.3%
Advisor Assignment	42.3%	-57.7%
SEVIS Linkage Rate	37.8%	-62.2%

Validation Logic: KPIs calculated as (compliant records / total records) per intake. Cross-validated Connect vs SEVIS counts to ensure no double-counting.

04 | I-20 RATE vs DEPOSIT RATE — TREND ANALYSIS



05 | DATA-DRIVEN RECOMMENDATIONS

Fix I-20 Processing Pipeline

CRITICAL

R1

Spring 2025 shows 0.8% issuance — near-total failure. Conduct root-cause audit of document processing workflows. Implement automated triggers when issuance drops below 50% threshold.

Deposit Completion Intervention Program

HIGH

R2

At 21.9%, deposit completion is the weakest compliance metric. Implement automated reminder sequences at 7, 14, and 30 days post-admission. Target Africa and Asia cohorts first — highest volume.

SEVIS Fee Payment Monitoring Dashboard

HIGH

R3

Build a dedicated alert layer in Looker Studio that flags students with I-20 issued but SEVIS fee unpaid after 15 days. Currently 45.3% unpaid — this is a DHS compliance breach risk.

Mandate Advisor Assignment at Application Stage

MEDIUM

R4

42.3% advisor assignment rate means over half of students navigate compliance alone. Assign advisors at application confirmation, not post-enrollment — reduces downstream compliance gaps.

SEVIS Linkage Automated Reconciliation

MEDIUM

R5

37.8% linkage rate means 62% of students are not verified in SEVIS. Run weekly automated reconciliation between Connect and SEVIS records. Escalate unlinked records after 30 days.

Seasonal Intake-Specific Compliance Playbooks

STANDARD

R6

Summer intake (1,219 students) has distinct compliance curve. Create intake-specific communication and deadline calendars. Do not apply Fall templates to Summer and Spring cohorts.

05 | RECOMMENDATION PRIORITY MATRIX

Recommendation	Target Metric	Impact	Effort	Priority
R1 — Fix I-20 Processing Pipeline	I-20 Rate (34.7%)	Critical	High	Immediate
R2 — Deposit Completion Program	Deposit Rate (21.9%)	High	Medium	30 Days
R3 — SEVIS Fee Payment Alert Layer	Fee Payment (54.7%)	High	Low	30 Days
R4 — Mandatory Advisor Assignment	Advisor Rate (42.3%)	Medium	Medium	60 Days
R5 — SEVIS Linkage Reconciliation	SEVIS Linkage (37.8%)	High	Low	30 Days
R6 — Intake-Specific Playbooks	All Metrics	Medium	Low	90 Days

06 | CONCLUSION & INSTITUTIONAL IMPACT

Analysis of 34,341 international student records reveals systemic compliance gaps across all five KPIs. The data shows all rates trending sharply downward year-over-year, with SEVIS linkage and deposit completion representing the highest institutional risk. These are not isolated data issues — they indicate structural process failures requiring immediate intervention.



Legal & DHS Risk

62.2% of students lack SEVIS linkage — institutional DHS audit exposure if not resolved within current semester.



Revenue Leakage

78.1% deposit non-completion means enrolled seats remain unfilled. Direct impact on international enrollment revenue targets.



Student Success Risk

Without I-20 and advisor assignment, students cannot maintain legal status — creating dropout risk and visa violations.



Data Infrastructure

Supabase + Looker Studio pipeline now operational and validated — provides a scalable foundation for real-time compliance monitoring.

This analytics project demonstrates end-to-end compliance intelligence — from raw data to actionable institutional strategy.

THANK YOU



SLU International Student Compliance Analytics
Week 4 Final Presentation | Team 5

34,341

Records

34.7%

I-20 Rate

21.9%

Deposit Rate

54.7%

Fee Payment

37.8%

SEVIS Link